

PDVM — Prediction Dependence & Variation Module

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: PGA™ — Prediction Governance Architecture

Parent Standard:

Collective Prediction Governance Standard (CPGS)

Operational Layer: Collective Prediction Governance Layer

Category:

Governance & Enforcement

Subcategory:

Prediction Governance

Type: Parent Standard Module

Governed Space:

Prediction Dependence & Variation

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AI-Readable: Yes

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Protection:

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Canonical Language:

English (UCL™)

Governed Space

Prediction Dependence & Variation

Canonical Definition

Prediction Dependence & Variation Module (PDVM) defines the governance conditions under which predictive contributions within a potential collective prediction are assessed for dependence, shared informational origin, common methodology, behavioral influence, feedback effects, correlation, convergence, divergence, and meaningful predictive variation.



PDVM determines whether multiple predictive outputs represent:
materially independent predictive judgments

or:

dependent expressions of the same underlying information, model, assumption, signal, process, or social influence.

It also governs whether observed similarity or disagreement among contributing predictions reflects meaningful variation or merely superficial numerical difference.

PDVM transforms:

Governed Predictive Source Diversity Profile + Contributing Predictions + Relevant Source Relationships + Predictive Conditions

into:

Governed Prediction Dependence & Variation Profile

Its foundational principle is:

Multiple predictions do not become independent merely because they were produced separately.

Operational Role

PDVM is the second internal governance space of the Collective Prediction Governance Standard (CPGS) .

PSDM establishes:

what predictive source diversity exists.

PDVM establishes:

how dependent or independent the resulting predictions actually are, and what variation exists among them.

The focused operational question is:

To what extent do the contributing predictions represent genuinely distinct predictive judgments rather than repeated, correlated, influenced, or structurally dependent expressions of the same predictive conditions?

The primary output is:

Governed Prediction Dependence & Variation Profile

Internal Module Flow

Predictive Source Diversity → Prediction Dependence & Variation → Prediction Aggregation → Predictive Influence & Concentration → Collective Prediction Representation

Current Position: Prediction Dependence & Variation

Prediction Dependence

Prediction Dependence exists where one predictive contribution is materially influenced by, derived from, constrained by, or informationally connected to another predictive contribution or to a shared underlying predictive source.

Prediction Dependence ≠ Prediction Similarity

Two predictions can be similar and independent.

Two predictions can differ while still being dependent.

Prediction Independence

Prediction Independence means that predictive contributions are sufficiently distinct in formation that one cannot reasonably be treated as merely reproducing or inheriting the material predictive structure of another.

Independence ≠ Isolation

Predictors do not need completely separate worlds to be meaningfully independent.

The governance question is whether shared relationships materially affect collective interpretation.

Multiple Predictors ≠ Independent Predictions

Separate Submission ≠ Independent Prediction

Submitting predictions separately does not establish independence.

Shared Prediction Basis

Predictions may depend upon the same underlying information.

Examples include:

the same government report; the same market data;

the same research paper; the same sensor network; the same proprietary dataset; the same expert source.

Shared Prediction Basis ≠ Independent Confirmation

Shared Source Dependency

A Shared Source Dependency exists where multiple predictions materially depend upon one upstream source.

Different Analysts + Same Upstream Source ≠ Fully Independent Predictive Information

Common Information Environment

Predictors may consume broadly the same information environment even where no single common source can be identified.

Common Information Environment ≠ Identical Prediction Basis

Dependence may exist at different levels.

Information Overlap

Information Overlap represents the degree to which predictive contributions rely upon materially overlapping information.

Partial Information Overlap

Predictions may share some information while also containing distinct inputs.

Partial Overlap ≠ Full Dependence

Information Dependence Gradient

Prediction dependence need not be binary.

It may range across states such as:

Low Dependence

Moderate Dependence

High Dependence

Dominant Shared Dependence

Dependence State $\neq$ Correlation Coefficient by Default

A numerical correlation measure may support analysis but does not fully describe structural dependence.

Methodological Dependence

Predictions may depend upon the same or substantially similar predictive methodology.

Same Methodology $\neq$ Same Prediction by Default

However, common methodology may reduce methodological diversity.

Methodological Dependence $\neq$ Information Dependence

These should remain distinct.

Model Dependence

Multiple AI predictions may be dependent because they use:

the same base model; the same model family; the same fine-tuning; the same retrieval pipeline; the same upstream model output; the same orchestration structure.

Different Model Interface $\neq$ Model Independence

Shared Model Family $\neq$ Automatic Full Dependence

Model-family relationship is a governance signal, not a complete independence determination.

Prompt Dependence

Multiple model outputs may be shaped by the same prompt or system instruction.

Different Outputs from Same Prompt $\neq$ Independent Predictive Origins by Default

Retrieval Dependence

AI systems may retrieve from the same information sources.

Different Models + Same Retrieval Corpus $\neq$ Full Information Independence

Agent Dependence

Autonomous agents may depend upon:

the same base model; the same memory; the same tool outputs; the same orchestration layer; the same upstream agent; one another.

Multiple Agents $\neq$ Independent Predictors

Human Dependence

Human predictors may depend upon:

the same briefing; the same institution; the same expert; the same dominant narrative; the same market price; one another.

Different People ≠ Independent Predictive Formation by Default

Institutional Dependence

Predictions produced by different individuals within one institution may reflect shared: research; data; assumptions; incentives; decision structures; methodologies.

Institutional Separation ≠ Information Independence

Cross-Institution Dependence

Different institutions may still depend upon common upstream information.

Multiple Institutions ≠ Independent Information Ecosystems

Prediction Influence

Prediction Influence occurs where one prediction affects the formation of another prediction.

Observed Prediction → Later Prediction

This relationship should remain distinguishable from independent predictive formation.

Feedback Contamination

Feedback Contamination occurs where contributors observe collective or prior predictive outputs before submitting their own contributions, creating dependence within the collective.

Observed Consensus → Contribution ≠ Independent Contribution by Default

Sequential Dependence

Predictions may be formed sequentially rather than simultaneously.

Later Prediction ≠ Independent Prediction if Earlier Predictions Were Observed

Prediction Cascade

A Prediction Cascade occurs where later contributors follow or adapt to earlier predictions.

Cascade ≠ Independent Consensus

Herding

Herding occurs where predictors align with prevailing predictive positions because of:

social pressure; reputational incentives; institutional pressure; market signals; perceived expertise; fear of being an outlier.

Agreement Caused by Herding ≠ Independent Predictive Agreement

Anchoring

Anchoring occurs where a prior probability, forecast, market price, or expert estimate materially influences subsequent predictive judgments.

Anchored Prediction $\neq$ Fully Independent Prediction

Market Price Anchoring

Prediction-market participants may use current price as an input into their own prediction.

This creates reflexive dependence.

Market Participants $\neq$ Independent Forecasting Population by Default

Consensus Dependence

Consensus itself may become an input to later predictions.

Consensus Observed Before Prediction $\neq$ Independent Consensus Formation

Authority Dependence

A predictor may defer to perceived authoritative sources.

Source Authority $\neq$ Independent Predictive Judgment

Information Cascade

An Information Cascade occurs where predictors infer information from the actions or predictions of others rather than from independently observed evidence.

Observed Behavior $\neq$ Independent Information

Outcome-Proximity Dependence

Several predictors may depend upon one outcome-proximate source.

Multiple Predictors + One Outcome-Informed Source $\neq$ Multiple Independent Informational Advantages

Predictor–Reality Proximity Dependence

Sources may occupy different organizational identities while ultimately depending upon the same decision-proximate or outcome-influencing actor.

Different Source Identity $\neq$ Different Reality Proximity Origin

Common Decision Source

A Common Decision Source exists where multiple predictors depend upon information originating from the same decision-making authority.

Multiple Predictions from One Decision Source $\neq$ Independent Outcome Knowledge

Coordination Dependence

Predictors may coordinate deliberately.

Coordinated Prediction ≠ Independent Prediction

Coordination may be legitimate, but it must not be misrepresented as independence.

Strategic Dependence

Predictors may adapt their contributions in response to expected behavior of others.

Strategic Diversity ≠ Independent Predictive Diversity

Incentive Dependence

Shared incentives may encourage correlated predictions.

Shared Incentive ≠ Shared Information by Default

But shared incentives may still affect predictive behavior.

Dependence Channel

A Dependence Channel is the mechanism through which predictive contributions become materially connected.

Relevant channels may include:

information; methodology; model; data; institutional structure; social influence; market signals; coordination; shared assumptions; decision proximity.

Dependence Channel ≠ Dependence Magnitude

The channel identifies how dependence occurs.

Its magnitude identifies how materially it affects interpretation.

Direct Dependence

A prediction is directly dependent where it explicitly uses another prediction or source output.

Indirect Dependence

A prediction may be indirectly dependent through an intermediary source.

Indirect Dependence ≠ Immaterial Dependence

Dependence Chain

Dependence may propagate through multiple stages.

Example:

Government Forecast → Analyst Report → AI Retrieval → User Forecast

Longer Dependence Chain ≠ Greater Independence

Distance from the original source does not necessarily remove dependence.

Dependence Network

Collective prediction structures may form networks of predictive dependence.

Prediction Count $\neq$ Independent Node Count

Effective Independent Contribution

An Effective Independent Contribution is a predictive contribution that adds materially distinct predictive information or judgment after relevant dependence is considered.

Nominal Contribution Count $\neq$ Effective Independent Contribution Count

Effective Independence

Effective Independence reflects the degree to which the collective contains materially distinct predictive contributions after dependence channels are accounted for.

Absolute Independence $\neq$ Required Condition

Collective prediction governance can function with partial dependence if the dependence is known and appropriately represented.

Dependence Transparency

Known dependence is generally more governable than hidden dependence.

Dependence Exists $\neq$ Collective Prediction Invalid by Default

Hidden Dependence

Hidden Dependence creates greater governance risk because collective strength may be overstated.

Known Dependence $\neq$ Hidden Dependence

Prediction Variation

Prediction Variation describes meaningful differences among predictive contributions.

Variation may occur in:

predicted outcome; probability; range; assumptions; methodology; horizon; boundary; feasibility state; confidence; scenario; timing.

Numerical Difference $\neq$ Meaningful Predictive Variation

Two different numbers may reflect the same underlying predictive logic.

Numerical Similarity $\neq$ Predictive Sameness

Two identical probabilities may arise from materially different reasoning.

Same Probability $\neq$ Same Prediction

Probability Variation

Variation in probability values should remain interpretable within PPUS governance.

Probability Dispersion

Probability Dispersion represents the distribution of probability estimates across contributors.

Low Dispersion $\neq$ High Independence

Low Dispersion $\neq$ High Predictive Quality

High Dispersion $\neq$ Poor Collective Quality by Default

Outcome Variation

Predictors may disagree on the actual predicted outcome.

Outcome Variation $\neq$ Probability Variation

Range Variation

Contributors may provide different ranges or intervals.

Range Overlap $\neq$ Predictive Agreement by Default

Assumption Variation

Predictions may differ because assumptions differ.

PAS governs individual assumptions.

PDVM preserves the existence of assumption-based variation relevant to collective interpretation.

Different Assumptions $\neq$ Independent Prediction by Default

Methodological Variation

Contributors may use different methodologies.

Methodological Variation $\neq$ Informational Independence

Feasibility Variation

Contributing predictions may assign different feasibility states or rely upon different operational feasibility conditions.

Feasibility Disagreement $\neq$ Probability Disagreement

Temporal Variation

Predictions formed at different times may differ because the information environment changed.

Prediction at Time A $\neq$ Prediction at Time B

Temporal Variation $\neq$ Source Diversity

Boundary Variation

Contributors may use different predictive boundaries.

Boundary Difference $\neq$ Legitimate Predictive Variation for Aggregation by Default

Horizon Variation

Predictions may use different time horizons.

Different Horizon $\neq$ Comparable Variation by Default

Variation Source

PDVM should preserve, where identifiable, why predictions differ.

Variation may result from:

different information; different assumptions; different methodology; different horizon; different boundary; different model; different operational knowledge; different uncertainty representation.

Variation Without Source Understanding ≠ Fully Governed Variation

Independent Convergence

Independent Convergence occurs where materially independent predictive contributions reach similar conclusions.

Independent Convergence ≠ Herded Consensus

This is one of the most important PDVM distinctions.

Dependent Convergence

Dependent Convergence occurs where similar predictions arise because contributors share the same predictive structure.

Agreement ≠ Independent Convergence

Independent Divergence

Independent predictors may legitimately disagree.

Independent Divergence ≠ Governance Failure

Dependent Divergence

Dependent sources may still produce different outputs because of:
model randomness; parameter variation; prompt variation; interpretation differences.

Different Outputs ≠ Independent Origins

Consensus Strength

Consensus should be interpreted together with dependence.

High Consensus + High Dependence ≠ Strong Independent Confirmation

Moderate Consensus + Low Dependence May Carry Different Meaning

PDVM does not itself determine predictive quality, but it preserves the structure necessary for legitimate interpretation.

Variation Concentration

Variation may be concentrated among particular source groups.

Minority Variation

A minority group may provide materially distinct predictive information.

Minority Variation ≠ Noise by Default

Outlier Variation

Outliers may arise from genuine independent information or methodological difference.

Outlier ≠ Invalid Contribution

False Diversity

False Diversity occurs where prediction outputs differ numerically but originate from substantially the same predictive structure.

Different Numbers ≠ Independent Predictive Diversity

False Consensus

False Consensus occurs where apparently broad agreement is materially produced by hidden dependence.

Broad Agreement ≠ Broad Independent Confirmation

Dependence Concentration

Dependence Concentration exists where many predictions ultimately rely upon one upstream source or dependence channel.

Many Downstream Predictions ≠ Many Upstream Information Origins

Single-Origin Dependence

A collective may contain numerous contributions but only one dominant predictive origin.

Collective Size ≠ Independent Information Capacity

Dependence Cluster

Predictions with materially shared dependence may be grouped into Dependence Clusters.

Dependence Cluster ≠ Single Prediction

The cluster represents structural relationship, not automatic merger.

Cross-Cluster Independence

Different clusters may still share upstream dependencies.

Different Clusters ≠ Independent Clusters by Default

Dependence Severity

Dependence may be classified where useful as:

Minor

Material

High

Dominant

Dependence Severity ≠ Predictive Error Risk by Default

PDVM governs structural dependence, not outcome correctness.

Dependence Uncertainty

In some cases, source relationships cannot be fully established.

Unknown Dependence ≠ Independence

Independence Claim

Where a collective mechanism claims independent prediction, the basis for that claim should remain governable.

Claimed Independence ≠ Demonstrated Independence

Independence State

PDVM may classify a contributing structure as:

Materially Independent

Partially Independent

Structurally Dependent

Highly Dependent

Dominantly Dependent

Independence Uncertain

Unable to Determine

Materially Independent

Contributions contain sufficiently distinct predictive formation to support meaningful independent interpretation.

Partially Independent

Some material independence exists alongside relevant shared dependence.

Structurally Dependent

Important dependence channels materially connect contributors.

Highly Dependent

A substantial portion of contributions share important predictive origins or influence channels.

Dominantly Dependent

The collective is largely driven by one or a small number of common predictive origins.

Independence Uncertain

Available information suggests possible dependence but cannot resolve its extent.

Unable to Determine

Available information is insufficient to establish dependence state.

Unable to Determine ≠ Independent

Variation State

PDVM may classify variation as:

Meaningfully Diverse

Moderately Varied

Highly Convergent

Highly Divergent

Dependence-Dominated

Variation Uncertain

Unable to Determine

Dependence-Dominated Variation

Variation may exist numerically while most contributions remain structurally dependent.

Variation State $\neq$ Predictive Quality State

Dependence-Adjusted Interpretation

Collective interpretation should consider predictive variation together with dependence.

Raw Dispersion $\neq$ Dependence-Adjusted Dispersion

Dependence and Aggregation

Prediction aggregation should not treat dependent contributions as though each represented fully independent predictive information.

Contribution Count $\neq$ Independent Information Count

Dependence $\neq$ Automatic Exclusion

Dependent predictions may still carry useful information.

The aggregation layer determines how contributions are composed.

Relationship to PSDM

PSDM governs predictive source diversity.

PDVM governs predictive dependence and variation.

Source Diversity $\neq$ Prediction Independence

Relationship to PBAS

PBAS governs individual Prediction Basis.

PDVM may identify shared or overlapping basis relationships without taking ownership of Prediction Basis governance.

Relationship to PAS

PAS governs Predictive Assumptions.

PDVM preserves assumption overlap or variation where relevant to dependence.

Relationship to PPUS

PPUS governs Probability & Uncertainty.

PDVM governs variation among governed probability representations without replacing PPUS.

Relationship to PFES

PFES governs Prediction Feasibility.

PDVM may preserve feasibility variation among contributing predictions.

Relationship to Predictor–Reality Proximity

Predictor–Reality Proximity may create shared outcome-proximate dependence where several predictors rely upon the same actor, decision source, or operational process.

High Proximity ≠ Independence

Relationship to Prediction Markets

Prediction-market participants may be dependent because they observe:

the current market price; prior trades; public consensus; shared news.

Market Participation ≠ Independent Forecasting

Relationship to AI Ensembles

AI ensemble diversity must distinguish:

model plurality; model-family dependence; shared retrieval; shared prompts; common upstream data.

Multiple Models ≠ Multiple Independent Predictions

Governance Requirements

PDVM requires that:

1. Prediction Dependence remain distinguishable from Prediction Similarity; 2. Prediction Independence remain distinguishable from isolation; 3. separate submissions not automatically establish independence; 4. shared Prediction Basis remain identifiable where material; 5. Shared Prediction Basis not be treated as independent confirmation; 6. common information environments remain governable; 7. partial overlap remain distinguishable from full dependence; 8. dependence states remain representable as gradients where appropriate; 9. structural dependence remain distinguishable from numerical correlation; 10. methodological dependence remain distinguishable from information dependence; 11. model dependence remain identifiable where AI systems contribute; 12. different interfaces not automatically establish model independence; 13. shared model-family relationships remain preservable; 14. prompt dependence remain representable where material; 15. retrieval dependence remain identifiable; 16. agent dependence remain identifiable; 17. multiple agents not be treated as independent by default; 18. human dependence remain assessable; 19. institutional dependence remain identifiable; 20. cross-institution dependence remain possible; 21. Prediction Influence remain distinguishable from independent formation; 22. feedback contamination remain identifiable; 23. sequential dependence remain governable; 24. Prediction Cascades remain distinguishable from independent consensus;
25. herding remain identifiable where material; 26. anchoring remain representable; 27. market-price anchoring remain distinguishable from independent judgment; 28. consensus dependence remain governable; 29. authority dependence remain identifiable where material; 30. information cascades remain distinguishable from independent information; 31. outcome-proximity dependence remain governable; 32. Predictor–Reality Proximity dependence remain identifiable where material; 33. common decision-source dependence remain representable; 34. coordinated prediction remain distinguishable from independent

prediction; 35. strategic dependence remain identifiable where material; 36. shared incentives remain distinguishable from shared information; 37. dependence channels remain identifiable; 38. dependence channels remain distinguishable from dependence magnitude; 39. direct and indirect dependence remain distinguishable; 40. indirect dependence not be treated as immaterial by default; 41. dependence chains remain traceable where material; 42. distance from source not be treated as independence; 43. dependence networks remain representable where useful; 44. nominal contribution count remain distinguishable from effective independent contribution count; 45. absolute independence not be required by default; 46. known dependence remain distinguishable from hidden dependence; 47. dependence not automatically invalidate a collective prediction; 48. Prediction Variation remain distinguishable across probability, outcome, assumptions, methodology, horizon, boundary, and feasibility; 49. numerical difference not automatically establish meaningful variation; 50. numerical similarity not establish predictive sameness; 51. Probability Dispersion remain distinguishable from independence; 52. low dispersion not automatically establish predictive quality; 53. high dispersion not automatically establish poor collective quality; 54. outcome variation remain distinguishable from probability variation; 55. assumption variation remain preservable; 56. methodological variation remain distinguishable from informational independence; 57. feasibility variation remain distinguishable from probability disagreement; 58. temporal variation remain distinguishable from source diversity; 59. horizon and boundary variation remain explicit; 60. sources of variation remain identifiable where possible; 61. Independent Convergence remain distinguishable from herded consensus; 62. Dependent Convergence remain identifiable; 63. Independent Divergence not be treated as governance failure; 64. different outputs not automatically establish independent origins; 65. consensus strength remain interpreted together with dependence; 66. minority variation not be treated as noise by default; 67. outliers not be invalidated solely because they differ; 68. False Diversity remain identifiable; 69. False Consensus remain identifiable; 70. Dependence Concentration remain governable; 71. many downstream predictions not be treated as many upstream information origins; 72. dependence clusters remain representable; 73. different clusters not automatically be treated as independent; 74. dependence severity remain distinguishable from predictive error risk; 75. unknown dependence not be interpreted as independence; 76. claimed independence remain distinguishable from demonstrated independence; 77. Independence Uncertain and Unable to Determine remain available states; 78. Dependence-Adjusted Interpretation remain possible; 79. dependent predictions not automatically be excluded; 80. Prediction Dependence & Variation remain distinct from aggregation and final Collective Prediction Representation.

Governance Logic

PDVM applies the following governance logic:

Governed Predictive Source Diversity Profile

Contributing Predictions

Relevant Prediction Basis Relationships

Relevant Methodological / Model / Agent / Institutional Relationships

↓

Prediction Relationships Identified

↓

Shared Information Sources Assessed

↓

Information Overlap Assessed

↓

Methodological Dependence Assessed

↓

Model / Agent Dependence Assessed where applicable

↓

Institutional / Organizational Dependence Assessed

↓

Sequential Influence Assessed

↓

Feedback Contamination Assessed

↓

Herding / Anchoring / Prediction Cascades Assessed

↓

Predictor–Reality Proximity Dependence Assessed where material

↓

Dependence Channels Identified

↓

Dependence Concentration Assessed

↓

Dependence Clusters Identified where useful

↓

Effective Independent Contribution Structure Established

↓

Prediction Variation Assessed

↓

Probability / Outcome / Assumption / Methodology / Horizon / Boundary Variation Distinguished

↓

Independent Convergence / Dependent Convergence Assessed

↓

Independent Divergence / Dependence-Dominated Variation Assessed

↓

False Diversity / False Consensus Assessed

↓

Independence State Established

↓

Variation State Established

↓

Historical Dependence & Variation State Preserved

↓

Governed Prediction Dependence & Variation Profile

↓

Prediction Aggregation

Where sufficient determination cannot be established:

Prediction Dependence & Variation Governance Gap → Source Relationship Clarification / Information Overlap Analysis / Dependence Qualification / Variation Qualification / Independence Uncertain / Unable to Determine

Minimum Implementation Framework

A minimum implementation of PDVM must establish:

A. Collective Prediction Reference

The candidate collective prediction or collective prediction process.

B. Contribution Identifier

A persistent identifier for each material predictive contribution.

C. Source Reference

The predictive source associated with each contribution.

D. Dependence Channel

Relevant information, methodology, model, institutional, social, market, or other dependence mechanism.

E. Shared Source / Shared Basis Reference

Where materially identifiable.

F. Dependence Direction

Where dependence is directional.

G. Dependence Magnitude / State

Where determinable.

H. Dependence Cluster

Where materially useful.

I. Effective Independent Contribution Structure

The governed representation of materially independent predictive contributions.

J. Variation Type

Probability, outcome, assumption, methodology, horizon, boundary, feasibility, or other relevant variation.

K. Variation State

The material state of predictive variation.

L. Convergence / Divergence State

Where relevant.

M. Independence State

At minimum:

Materially Independent / Partially Independent / Structurally Dependent / Highly Dependent / Dominantly Dependent / Independence Uncertain / Unable to Determine

N. Dependence Qualification

Any material limitation or unresolved dependency.

O. Historical Dependence & Variation Version

The reconstructable state applicable at the relevant lifecycle point.

Governance Outputs

PDVM may produce:

Governed Prediction Dependence & Variation Profile; Prediction Relationship Record; Information Overlap Record; Shared Prediction Basis Relationship; Methodological Dependence Record; Model Dependence Record; Agent Dependence Record;

Institutional Dependence Record; Sequential Dependence Record; Feedback Contamination Record; Prediction Cascade Record; Herding Condition Record; Anchoring Record; Market Price Anchoring Record; Predictor–Reality Proximity Dependence Record; Common Decision Source Record; Dependence Channel Register; Direct Dependence Record; Indirect Dependence Record; Dependence Chain; Dependence Network; Dependence Concentration Record; Dependence Cluster; Effective Independent Contribution Structure; Independence State; Dependence Severity State; Dependence Uncertainty Record; Prediction Variation Profile; Probability Dispersion Record; Outcome Variation Record; Assumption Variation Record; Methodological Variation Record; Feasibility Variation Record; Horizon Variation Record; Boundary Variation Record; Independent Convergence Record; Dependent Convergence Record; Independent Divergence Record; False Diversity Record; False Consensus Record; Variation State; Prediction Dependence & Variation Governance Gap; historical dependence state; historical variation state; Dependence & Variation Version.

The primary governed output is:

Governed Prediction Dependence & Variation Profile

which becomes an input to:

Prediction Aggregation.

Critical Distinctions

Multiple Predictions Do Not Become Independent Merely Because They Were Produced Separately

Prediction Dependence $\neq$ Prediction Similarity

Independence $\neq$ Isolation

Multiple Predictors $\neq$ Independent Predictions

Separate Submission $\neq$ Independent Prediction

Shared Prediction Basis $\neq$ Independent Confirmation

Different Analysts + Same Upstream Source $\neq$ Fully Independent Predictive Information

Partial Overlap $\neq$ Full Dependence

Dependence State $\neq$ Correlation Coefficient by Default

Methodological Dependence $\neq$ Information Dependence

Different Model Interface $\neq$ Model Independence

Shared Model Family $\neq$ Automatic Full Dependence

Different Outputs from Same Prompt $\neq$ Independent Predictive Origins by Default

Different Models + Same Retrieval Corpus $\neq$ Full Information Independence

Multiple Agents ≠ Independent Predictors

Different People ≠ Independent Predictive Formation by Default

Institutional Separation ≠ Information Independence

Multiple Institutions ≠ Independent Information Ecosystems

Observed Prediction → Later Prediction ≠ Independent Formation by Default

Observed Consensus → Contribution ≠ Independent Contribution by Default

Cascade ≠ Independent Consensus

Agreement Caused by Herding ≠ Independent Predictive Agreement

Anchored Prediction ≠ Fully Independent Prediction

Market Participants ≠ Independent Forecasting Population by Default

Consensus Observed Before Prediction ≠ Independent Consensus Formation

Observed Behavior ≠ Independent Information

Multiple Predictors + One Outcome-Informed Source ≠ Multiple Independent Informational Advantages

Different Source Identity ≠ Different Reality Proximity Origin

Multiple Predictions from One Decision Source ≠ Independent Outcome Knowledge

Coordinated Prediction ≠ Independent Prediction

Dependence Channel ≠ Dependence Magnitude

Indirect Dependence ≠ Immaterial Dependence

Longer Dependence Chain ≠ Greater Independence

Prediction Count ≠ Independent Node Count

Nominal Contribution Count ≠ Effective Independent Contribution Count

Absolute Independence ≠ Required Condition

Dependence Exists ≠ Collective Prediction Invalid by Default

Known Dependence ≠ Hidden Dependence

Numerical Difference ≠ Meaningful Predictive Variation

Numerical Similarity ≠ Predictive Sameness

Same Probability ≠ Same Prediction

Low Dispersion ≠ High Independence

Low Dispersion ≠ High Predictive Quality

High Dispersion ≠ Poor Collective Quality by Default

Outcome Variation ≠ Probability Variation

Different Assumptions ≠ Independent Prediction by Default

Methodological Variation ≠ Informational Independence

Feasibility Disagreement ≠ Probability Disagreement

Prediction at Time A ≠ Prediction at Time B

Temporal Variation ≠ Source Diversity

Different Horizon ≠ Comparable Variation by Default

Variation Without Source Understanding $\neq$ Fully Governed Variation

Independent Convergence $\neq$ Herded Consensus

Agreement $\neq$ Independent Convergence

Independent Divergence $\neq$ Governance Failure

Different Outputs $\neq$ Independent Origins

High Consensus + High Dependence $\neq$ Strong Independent Confirmation

Minority Variation $\neq$ Noise by Default

Outlier $\neq$ Invalid Contribution

Different Numbers $\neq$ Independent Predictive Diversity

Broad Agreement $\neq$ Broad Independent Confirmation

Many Downstream Predictions $\neq$ Many Upstream Information Origins

Collective Size $\neq$ Independent Information Capacity

Different Clusters $\neq$ Independent Clusters by Default

Dependence Severity $\neq$ Predictive Error Risk by Default

Unknown Dependence $\neq$ Independence

Claimed Independence $\neq$ Demonstrated Independence

Unable to Determine $\neq$ Independent

Variation State $\neq$ Predictive Quality State

Raw Dispersion $\neq$ Dependence-Adjusted Dispersion

Contribution Count $\neq$ Independent Information Count

Dependence $\neq$ Automatic Exclusion

Source Diversity $\neq$ Prediction Independence

High Proximity $\neq$ Independence

Market Participation $\neq$ Independent Forecasting

Multiple Models $\neq$ Multiple Independent Predictions

Enterprise Application

PDVM enables organizations to determine whether apparent collective predictive strength comes from genuinely distinct judgments or from repeated versions of the same predictive structure.

This is particularly important for:

prediction markets; expert panels; AI ensembles; multi-agent forecasting; institutional forecasting; crowdsourced prediction systems; financial forecasting; Human-AI prediction systems.

Without dependence governance, a system may count hundreds of predictions while effectively receiving only a handful of independent predictive signals.

PDVM makes that difference explicit before aggregation occurs.

Use Case 1 — 500 Analysts Appear to Confirm the Same Forecast

A financial platform collects forecasts from:

500 analysts

about whether a recession will begin within twelve months.

More than 80% of analysts converge around a similar probability range.

The platform describes this as strong collective confirmation.

PDVM examines the predictive dependence structure.

It finds that a large majority of analysts rely heavily upon:

the same central-bank projections; the same macroeconomic datasets; a small number of major economic research providers.

Many analysts also observe consensus forecasts before publishing their own estimates.

The critical governance question is:

Does agreement among hundreds of analysts represent hundreds of independent predictive judgments when their information sources and consensus signals are heavily shared?

PDVM identifies:

Shared Prediction Basis

Feedback Contamination

and:

Dependence Concentration

The collective may still contain useful information, but its agreement cannot legitimately be interpreted as 500 independent confirmations.

Broad Agreement $\neq$ Broad Independent Confirmation

Use Case 2 — Five AI Models Disagree, but the Diversity Is Mostly Superficial

An organization uses five AI systems to forecast whether a major infrastructure project will be completed on schedule.

Their predictions are:

58% · 63% · 67% · 71% · 76%

Management interprets the numerical spread as evidence of diverse predictive reasoning.

PDVM examines the systems.

All five retrieve from the same project database and the same public reporting corpus.

Three belong to the same model family, and all receive nearly identical forecasting instructions.

The critical governance question is:

Does numerical variation across several AI outputs establish meaningful predictive diversity when the systems share the same information and much of the same predictive structure?

PDVM classifies the variation as materially affected by dependence and prevents the numerical spread from being treated as independent information diversity.

Numerical Difference $\neq$ Meaningful Predictive Variation

Different Outputs $\neq$ Independent Origins

Canonical Boundary

PDVM governs the identification, characterization, qualification, and historical preservation of dependence and meaningful variation among predictive contributions within a potential collective prediction. It governs shared information, shared Prediction Basis relationships, methodological dependence, model and agent dependence, institutional dependence, sequential influence, feedback contamination, anchoring, herding, prediction cascades, Predictor–Reality Proximity dependence, dependence channels, dependence concentration, effective independent contribution structure, convergence, divergence, dispersion, False Diversity, and False Consensus. It does not govern Predictive Source Diversity itself, establish individual Prediction Basis, determine individual Prediction Probability, aggregate predictions, assign predictive influence, determine final Collective Prediction legitimacy, govern Prediction Communication & Reliance, or assess predictive performance.

PDVM governs:

how independent, dependent, convergent, divergent, and meaningfully varied the contributing predictions actually are.

It does not determine:

how those predictions should ultimately be aggregated into a collective prediction.

Canonical Closing Statement

A collective can look diverse because its outputs differ while still thinking through the same underlying source.

PDVM exposes the information links, model relationships, feedback effects, cascades, and shared predictive structures that determine whether multiple predictions truly contribute independent judgment.

Collective strength is not measured by how many predictions agree or disagree — it begins with understanding how independently those predictions were actually formed.

It

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